Debanjan Shil

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Professional Summary

Machine Learning Engineer specializing in Personalization and Recommendation (PnR) systems. Expert in architecting scalable ranking and semantic retrieval pipelines that bridge high-dimensional data with real-time user experiences. Proven track record in building "Honest AI" systems that utilize uncertainty quantification and Gaussian Processes to optimize model refresh strategies and mitigate hallucination risk in production-grade discovery engines.

Education

M.Tech in Data Science	Amrita Vishwa Vidyapeetham, Bengaluru	Expected Jun 2026
B.Tech in Civil Engineering	Maulana Abul Kalam Azad Univ. of Tech, Kalyani	Jun 2023

Projects

Scalable Discovery & Ranking Infrastructure (Project Anāhitā / SatelliteRL) Nov 2025 – Present

Python, PyTorch, LangChain, Vector Similarity, and Automated Model Lifecycle.

- Architected a semantic retrieval system utilizing high-dimensional embeddings to automate large-scale entity discovery, mirroring streaming-industry recommendation architectures.
- Developed an "Honest AI" ranking framework using Gaussian Process uncertainty to optimize model refresh cycles and ensure high-integrity content discovery.
- Integrated LLM-driven AI agents (LangChain) to automate complex metadata enrichment pipelines, significantly accelerating the model experimentation and reproducibility lifecycle.
- Optimized large-scale model training and inference loops, leveraging distributed computing to handle multi-modal features and semantic search workflows.

High-Traffic Production Engineering (Farmers for Forests) May 2026

FastAPI, Docker, SAHI, Detectron2, and Performance Monitoring.

- Engineered a real-time, production-grade inference pipeline for high-resolution data, utilizing SAHI (Tiled Inference) to reduce latency by 97.5% for millions of data points.
- Developed a "Gatekeeper" validation layer to cross-reference model ranking outputs against historical ground truth, ensuring 100% data reliability for predictive scoring.
- Established CI/CD and automated evaluation (Evals) workflows to support A/B testing, continuous learning loops, and rapid model iteration in live production environments.
- Mapped complex spatial datasets into scalable 2D-to-temporal structures using computational geometry to identify behavioral anomalies and ranking bias.

Spatio-Temporal Pattern Recognition (Master's Dissertation) Jun 2026

Graph Normalizing Flows (STG-NF), YOLO-pose, and Feature Stores.

- Architected a vision pipeline integrating STG-NF for behavioral pattern recognition, utilizing graph structures for high-precision entity tracking and ranking.
- Optimized real-time signal processing for 7,400+ data tensors, implementing custom filtering to improve signal-to-noise ratios in high-traffic streaming environments.

Technical Skills

- **ML Personalization:** Ranking Systems, User Embeddings, Semantic Search, NDCG, AUC, MAP, Re-ranking, Vector Similarity (FAISS), Transformers, PyTorch, TensorFlow.
- **Infrastructure & MLOps:** Databricks, Spark, MLflow, Feature Stores, Experiment Tracking, A/B Testing Design, Docker, CI/CD, FastAPI.
- **Generative AI & Agents:** LangChain, Agentic Orchestration, RAG, Prompt Engineering, LLM Evaluations (Evals), Reinforcement Learning (MARL).
- **Core Engineering:** Python (OOP), SQL, ETL Pipeline Automation, Git, Unit Testing (PyTest), AWS/GCP Environments.

Open Source & Certifications

Contributor to *Project Anāhitā* (Geospatial Discovery) | IBM Data Science Professional | AWS Machine Learning Foundation